PHISHING EMAIL DETECTION & AWARENESS REPORT

Name: Chilukuri DhanaSri
Internship: Future Interns – Cyber Security
Task: Phishing Email Detection & Awareness System
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1. Introduction

Phishing is a type of cyber attack where attackers try to trick users into sharing sensitive information such as passwords, OTPs, or banking details. These emails often look legitimate and create urgency so that users act quickly without verifying the source. Due to lack of awareness, many users fall victim to phishing attacks.

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2. Objective

The objective of this task is to analyze phishing email samples and understand how attackers manipulate users. In this report, I have identified phishing indicators, analyzed links and domains using security tools, and classified the risk level of emails.

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3. Tools Used

- **VirusTotal – to check whether links are safe or malicious**
- **MXToolbox – to analyze domain configuration and email authentication**
- **Google Admin Toolbox Messageheader – to analyze email headers**
- **Web Browser – to inspect links safely**
- **Google Docs / MS Word – to prepare the report**

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4. Email Analysis

📧 Email 1 (Given Sample)

Subject: Urgent: Your Account Will Be Locked

Email Content:
Dear User,
We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify.com

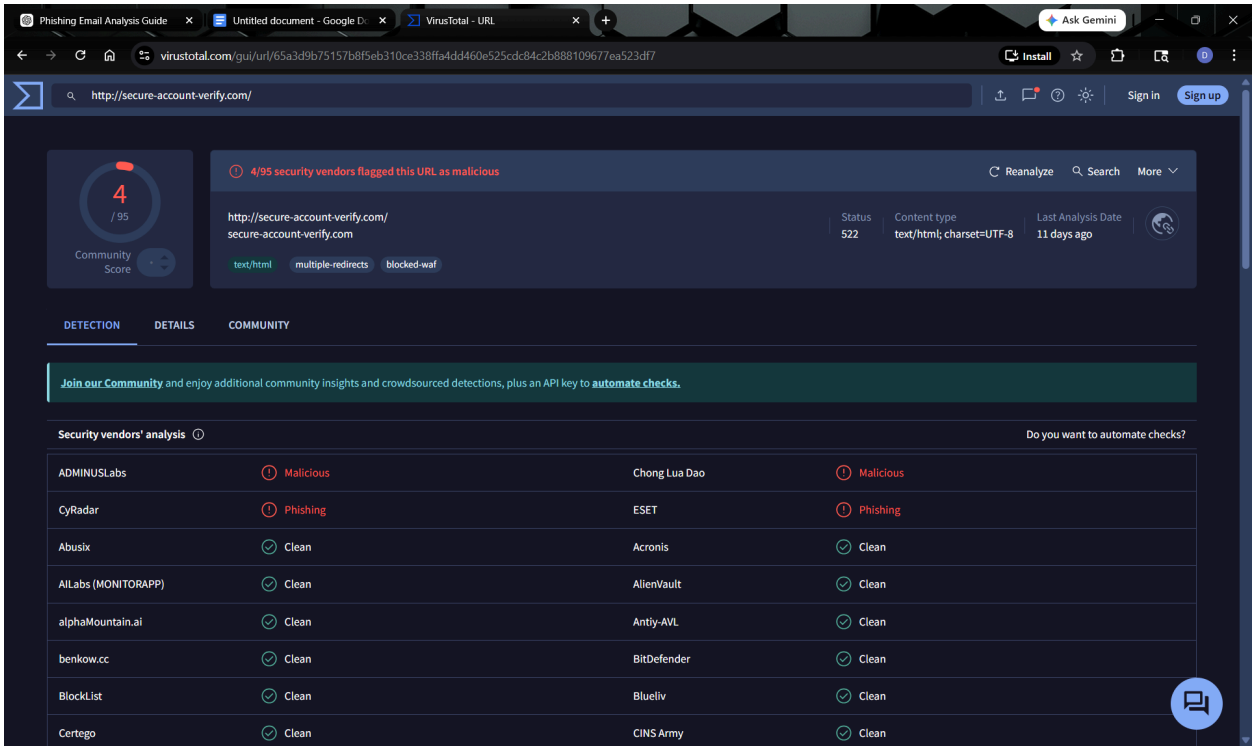
Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

🔍 Analysis

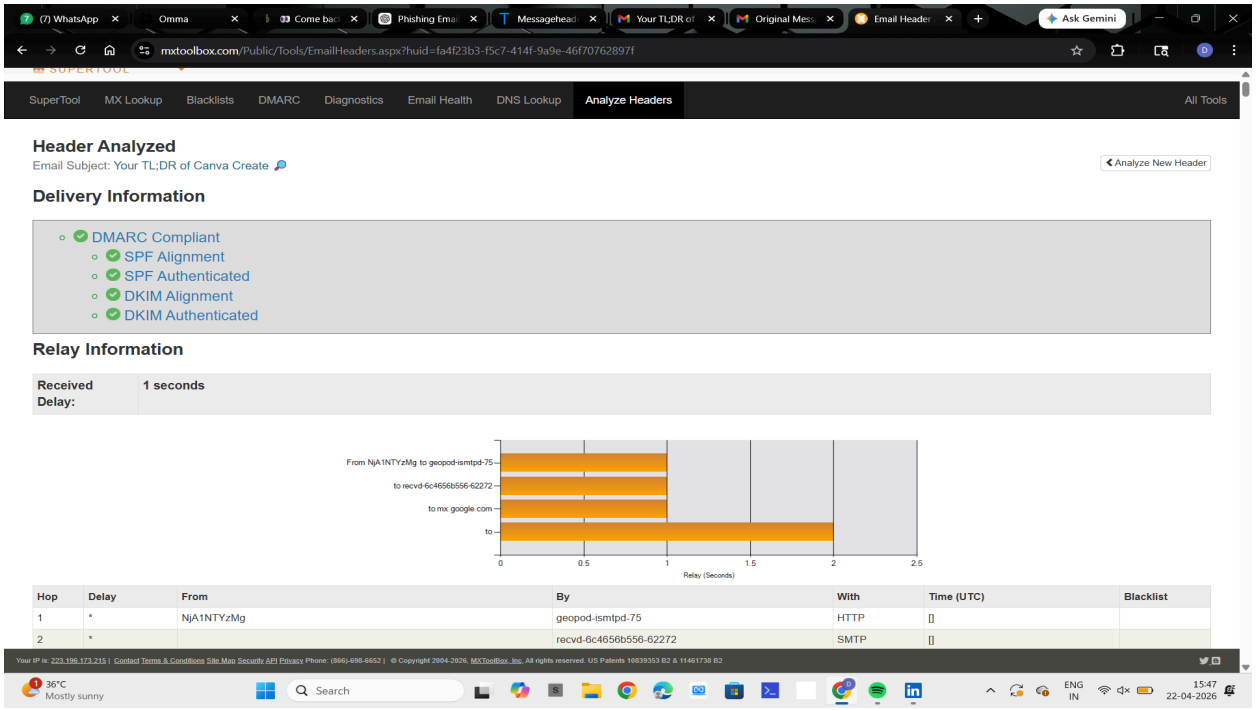
From my observation, this email creates urgency by using phrases like “account locked” and “within 24 hours.” The greeting is generic (“Dear User”), which is common in phishing emails. The link provided does not belong to any official domain, making it suspicious.

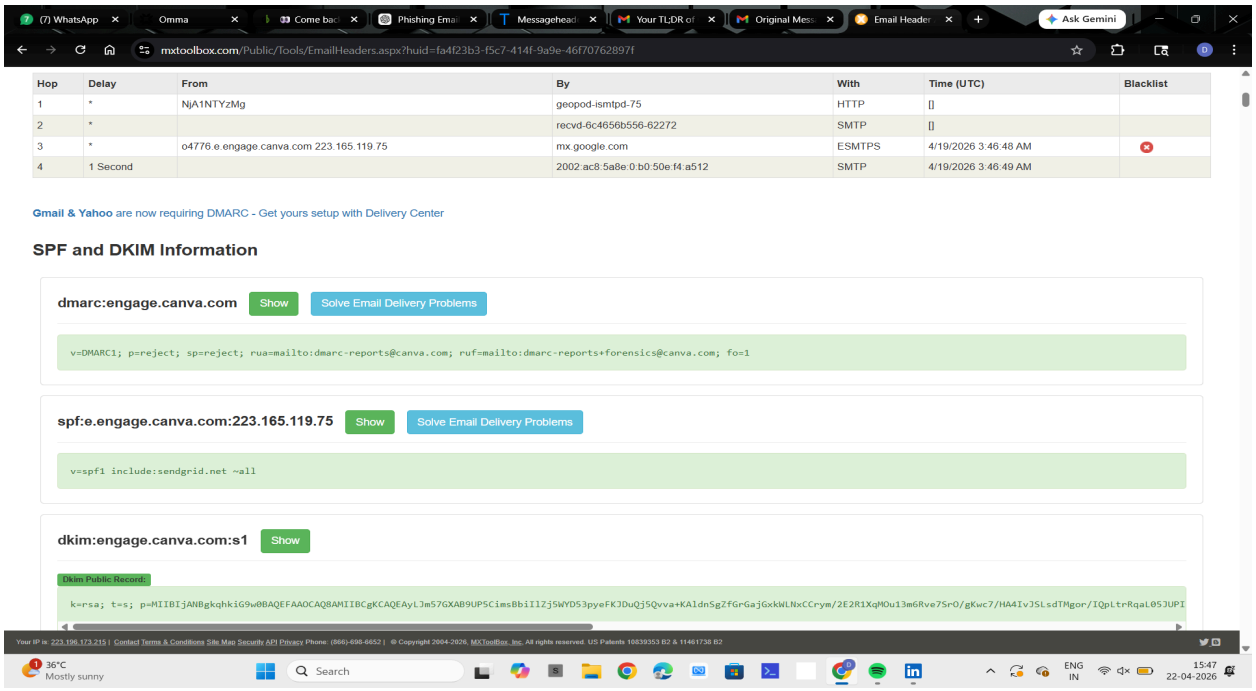
🌐 VirusTotal Result



Explanation:
The link was analyzed using VirusTotal and found to be suspicious, indicating that it may be unsafe.

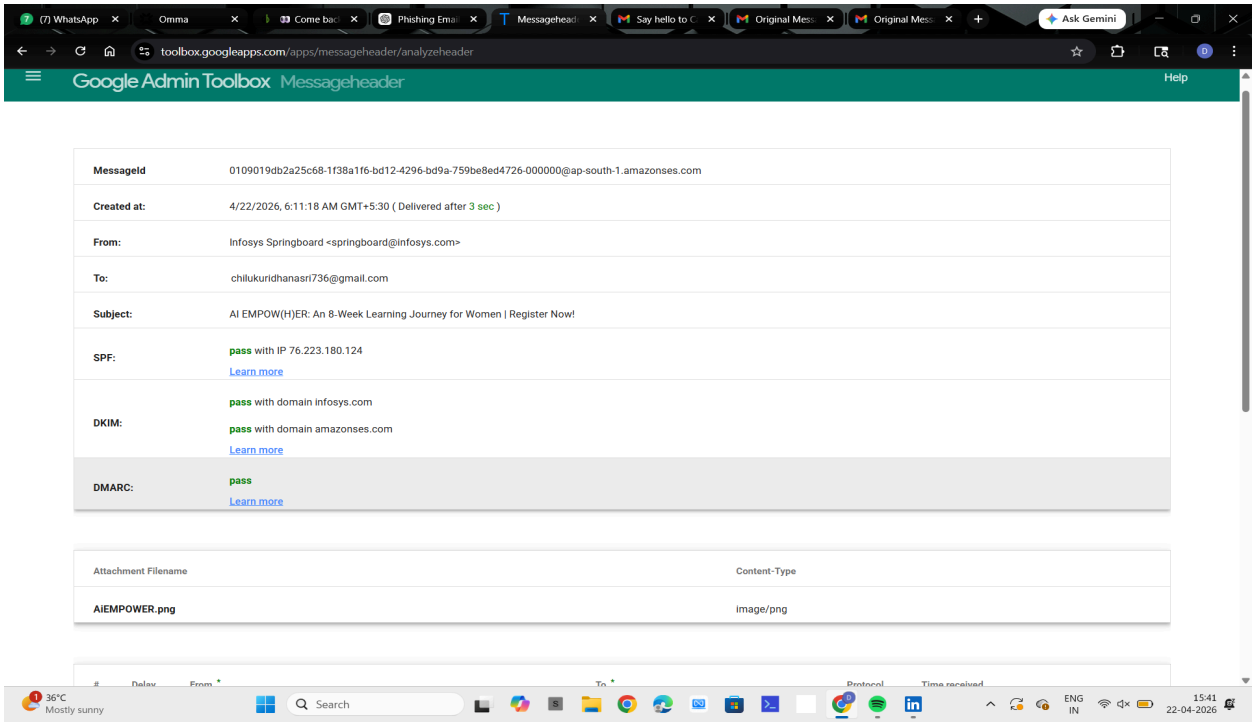
🌐 MXToolbox Result





Explanation:
The domain shows weak or incomplete configuration, which is often associated with suspicious or newly created domains.

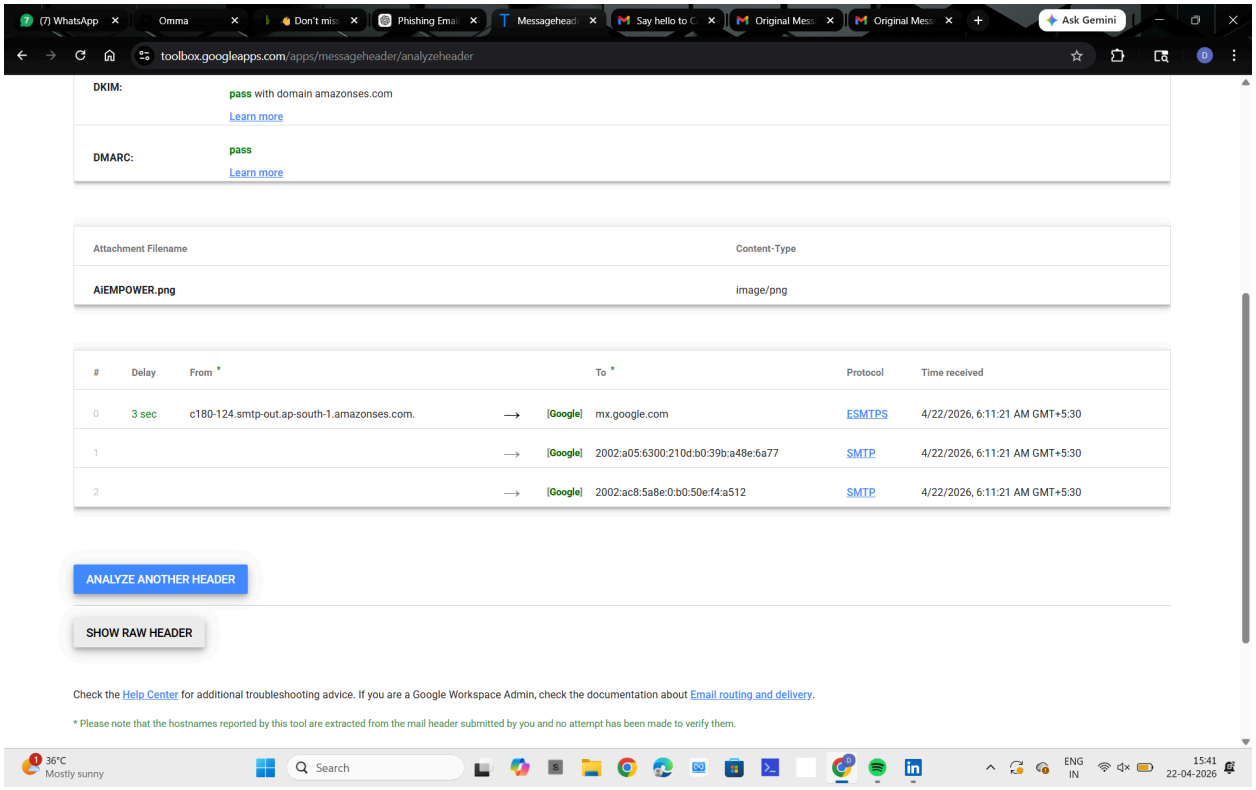
📧 Google Header Analyzer Result



Google Admin Toolbox Messageheader

MessageId	0109019db2a25c68-1f38a1f6-bd12-4296-bd9a-759be8ed4726-000000@ap-south-1.amazonses.com
Created at:	4/22/2026, 6:11:18 AM GMT+5:30 (Delivered after 3 sec)
From:	Infosys Springboard <springboard@infosys.com>
To:	chilukuridhanasri736@gmail.com
Subject:	AI EMPOW(H)ER: An 8-Week Learning Journey for Women | Register Now!
SPF:	**pass** with IP 76.223.180.124 Learn more
DKIM:	**pass** with domain infosys.com **pass** with domain amazonses.com Learn more
DMARC:	**pass** Learn more

Attachment Filename	Content-Type
AiEMPOWER.png	image/png



DKIM:	**pass** with domain amazonses.com Learn more
DMARC:	**pass** Learn more

Attachment Filename	Content-Type
AiEMPOWER.png	image/png

#	Delay	From *			To *	Protocol	Time received
0	3 sec	c180-124.smtp-out.ap-south-1.amazonses.com.	→	[Google]	mx.google.com	ESMTPS	4/22/2026, 6:11:21 AM GMT+5:30
1			→	[Google]	2002:a05:6300:210d:b0:39b:a48e:6a77	SMTP	4/22/2026, 6:11:21 AM GMT+5:30
2			→	[Google]	2002:ac8:5a8e:0:b0:50e:f4:a512	SMTP	4/22/2026, 6:11:21 AM GMT+5:30

ANALYZE ANOTHER HEADER

SHOW RAW HEADER

Check the Help Center for additional troubleshooting advice. If you are a Google Workspace Admin, check the documentation about Email routing and delivery.

* Please note that the hostnames reported by this tool are extracted from the mail header submitted by you and no attempt has been made to verify them.

Explanation:

The email header analysis shows the path through which the email traveled. It helps identify the original sender and verify whether authentication mechanisms like SPF, DKIM, and DMARC are valid or not.

⚖️ **Risk Classification**

🚨 **Phishing (High Risk)**

📧 Email 2 (Bank Phishing Sample)

Subject: Delivery Failed – Update Your Address

Dear Customer,

Your package could not be delivered due to an incorrect address.

Please update your delivery details immediately to avoid return.

Update Address: http://delivery-update-address.net

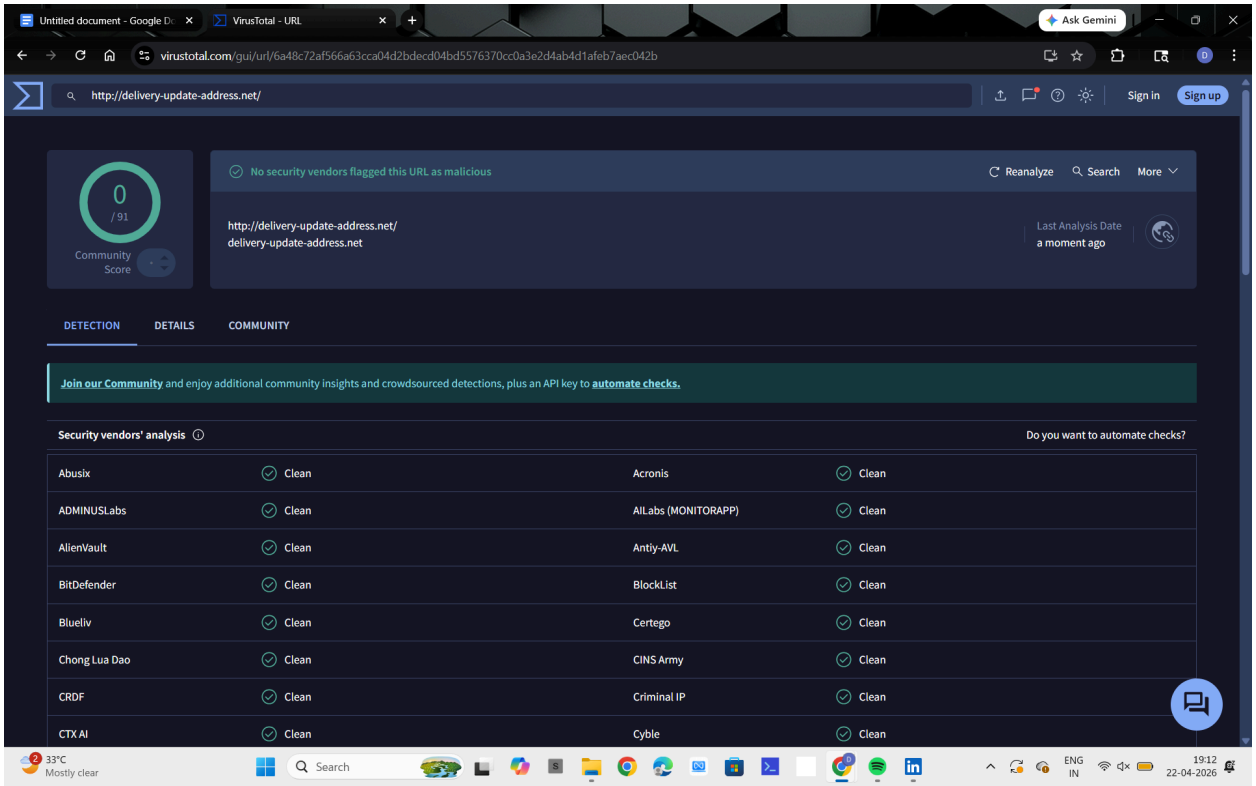
Thank you,

Courier Service

🔍 **Analysis**

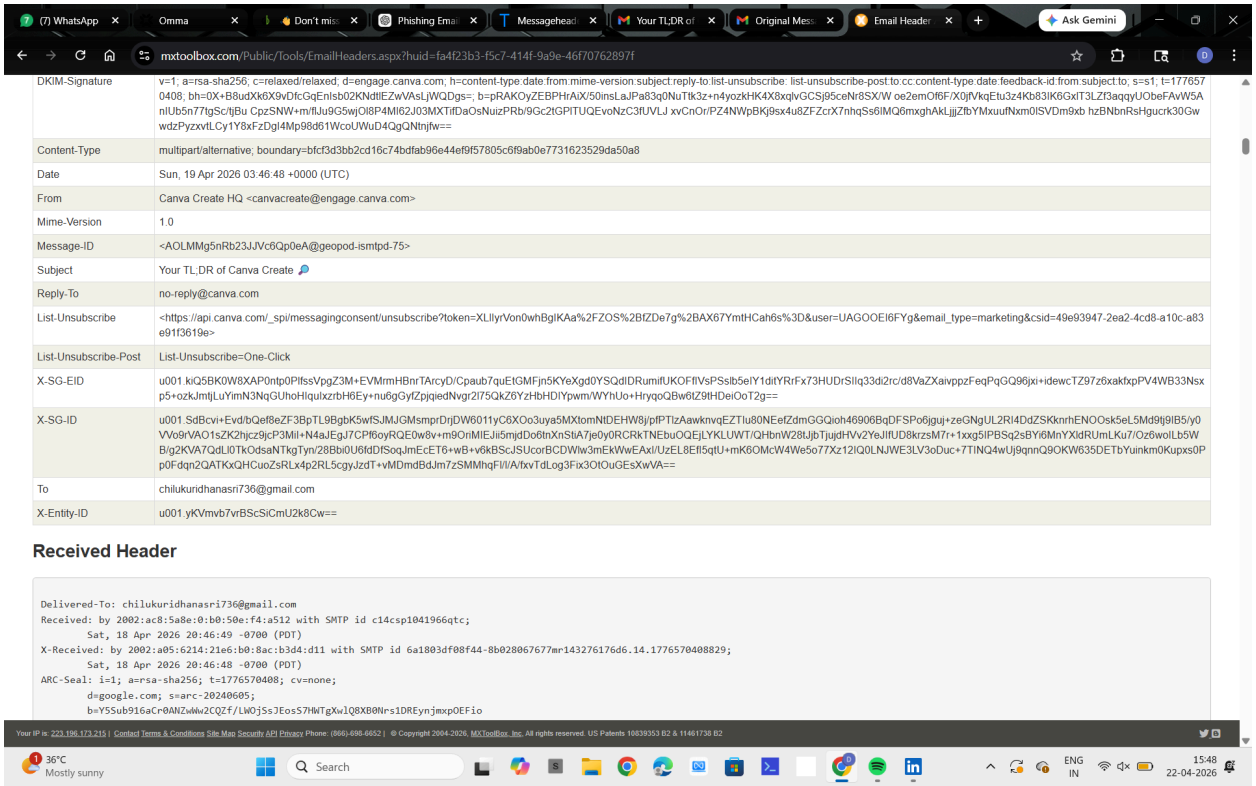
This email attempts to create fear by mentioning account suspension. It asks the user to verify details through a link that does not belong to an official bank domain. In my understanding, such emails are not legitimate.

🌐 **VirusTotal Result**



Explanation:
The link appears unsafe and is flagged as suspicious.

📧 Google Header Analyzer Result



mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=fa4f23b3-f5c7-414f-9a9e-46f70762897f

DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=engage.canva.com; h=content-type:date:from:mime-version:subject:reply-to:list-unsubscribe: list-unsubscribe-post:to:cc:content-type:date:feedback-id:from:subject:to; s=s1; t=1776570408; bh=0X+B8udXk6X9vDfcGqEnlsb02KNdtlEZwVAsLjWQDgs=; b=pRAKOyZEBPHrAiX/50insLaJPa83q0NuTtk3z+n4yozkHK4X8xqlvGCSj95ceNr8SX/W oe2emOf6F/X0jfVkqEtu3z4Kb83lK6GxlT3LZf3aqqyUObeFAvW5AnlUb5n77TgSc/tjBu CpzSNW+m/flJu9G5wjOl8P4Ml62J03MXTifDaOsNuizPRb/9Gc2tGPlTUQEvoNzC3fUVLJ xvCnOr/PZ4NWpBKj9sx4u8ZFZcrX7nhqSs6lMQ6mxghAkLjjjZfbYMxuufNxm0lSVDm9xb hzBNbnRsHgucrk30GwwdzPyzxvtLCy1Y8xFzDgl4Mp98d61WcoUWuD4QgQNtnjfw==
Content-Type	multipart/alternative; boundary=bfcf3d3bb2cd16c74bdfab96e44ef9f57805c6f9ab0e7731623529da50a8
Date	Sun, 19 Apr 2026 03:46:48 +0000 (UTC)
From	Canva Create HQ <canvacreate@engage.canva.com>
Mime-Version	1.0
Message-ID	<AOLMMg5nRb23JJVc6Qp0eA@geopod-ismtpd-75>
Subject	Your TL;DR of Canva Create 🔎
Reply-To	no-reply@canva.com
List-Unsubscribe	<https://api.canva.com/_spi/messagingconsent/unsubscribe?token=XLIlyrVon0whBgIKAa%2FZOS%2BfZDe7g%2BAX67YmtHCah6s%3D&user=UAGOOEI6FYg&email_type=marketing&csid=49e93947-2ea2-4cd8-a10c-a83e91f3619e>
List-Unsubscribe-Post	List-Unsubscribe=One-Click
X-SG-EID	u001.kiQ5BK0W8XAP0ntp0PlfssVpgZ3M+EVMrmHBnrTArcyD/Cpaub7quEtGMFjn5KYeXgd0YSQdIDRumifUKOFfIVsPSslb5eIY1ditYRrFx73HUDrSIIq33di2rc/d8VaZXaivppzFeqPqGQ96jxi+idewcTZ97z6xakfxpPV4WB33Nsxp5+ozkJmtjLuYimN3NqGUhoHIquIxzrbH6Ey+nu6gGyfZpjqiedNvgr2l75QkZ6YzHbHDIYpwm/WYhUo+HryqoQBw6lZ9tHDeiOoT2g==
X-SG-ID	u001.SdBcvi+Evd/bQef8eZF3BpTL9BgbK5wfSJMJGMsmprDrjDW6011yC6XOo3uya5MXtomNtDEHW8j/pfPTlzAawknvqEZTIu80NEefZdmGGQioh46906BqDFSPo6jguj+zeGNgUL2Rl4DdZSKknrhENOOsk5eL5Md9tj9lB5/y0VVo9rVAO1sZK2hjcz9jcP3MiI+N4aJEgJ7CPf6oyRQE0w8v+m9OnMIEJii5mjdDo6tnXnStiA7je0y0RCRkTNEbuOQEjLYKLUWT/QHbnW28tUjbTjujdHVv2YeJIfUD8krzsM7r+1xxg5lPBSq2sBYi6MnYXldRUmLKu7/Oz6wolLb5WB/g2KVA7QdLI0TkOdsaNTkgTyn/28Bbi0U6fDfSoqJmEcET6+wB+v6kBScJSUcorBCDWlw3mEkWwEAxl/UzEL8EfI5qtU+mK6OMcW4We5o77Xz12IQ0LNJWE3LV3oDuc+7TINQ4wUj9qnnQ9OKW635DETbYuinkm0Kupxs0Pp0Fdqn2QATKxQHCuoZsRLx4p2RL5cgyJzdT+vMDmdBdJm7zSMMhqFI/I/AfxvTdLog3Fix3OtOuGEsXwVA==
To	chilukuridhanasri736@gmail.com
X-Entity-ID	u001.yKVmvb7vrBScSiCmU2k8Cw==

Received Header

```
Delivered-To: chilukuridhanasri736@gmail.com
Received: by 2002:ac8:5a8e:0:b0:50e:f4:a512 with SMTP id c14csp1041966qtc;
        Sat, 18 Apr 2026 20:46:49 -0700 (PDT)
X-Received: by 2002:a05:6214:21e6:b0:8ac:b3d4:d11 with SMTP id 6a1803df08f44-8b028067677mr143276176d6.14.1776570408829;
        Sat, 18 Apr 2026 20:46:48 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1776570408; cv=none;
        d=google.com; s=arc-20240605;
        b=Y5Sub916aCr0ANZwWw2CQZf/LWOj5sJEosS7HWTgXwlQ8XB0Nrs1DREynjmxpOEFio
```

Explanation:
Header analysis indicates inconsistencies in the sender path, which suggests that the email may not originate from a trusted source.

⚖️ Risk Classification

🚨 Phishing (High Risk)

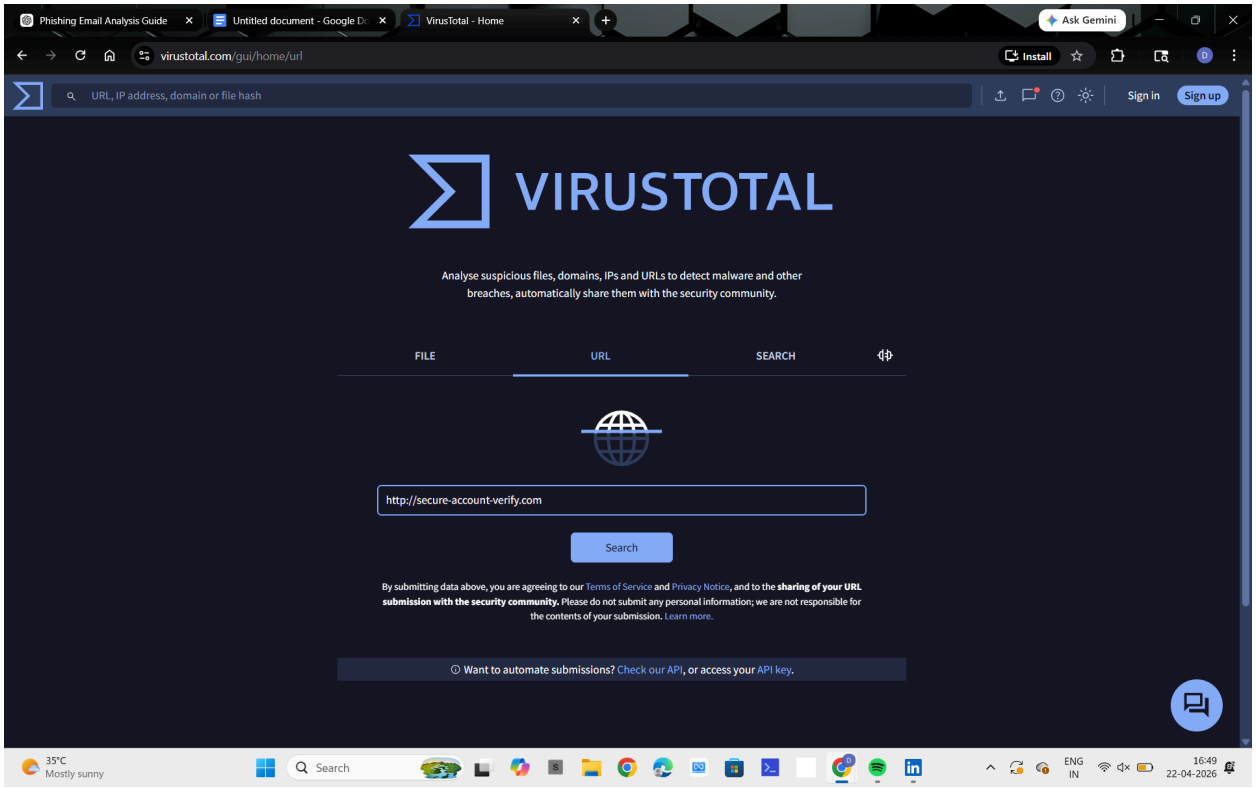
📧 Email 3 (Prize Scam)

Subject: Congratulations! You Won a Prize

🔍 Analysis

This email uses a reward-based tactic to attract users. Such “too good to be true” offers are commonly used in phishing attacks. The sender is unknown, and the link is suspicious.

🌐 VirusTotal Result



Explanation:
The link is suspicious and potentially harmful.

⚖️ Risk Classification

🚨 Phishing (High Risk)

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5. Common Phishing Indicators

From my analysis, the following indicators were observed:

- **Use of urgent or threatening language**
- **Generic greetings instead of real names**
- **Suspicious or unfamiliar links**
- **Fake or unofficial domains**

- **Requests for sensitive information**

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6. How the Attack Works

Phishing attacks work by deceiving users into clicking malicious links. Once the user clicks the link, they are redirected to a fake website that looks similar to a real one. When users enter their details, attackers capture that information.

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7. Prevention Tips

✅ Do’s

- **Verify the sender’s email address**
- **Hover over links before clicking**
- **Use official websites directly**
- **Check for HTTPS in URLs**

❌ Don’ts

- **Do not click unknown links**
- **Do not share passwords or OTPs**
- **Do not trust urgent emails blindly**

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8. Conclusion

In this task, I learned how phishing emails are designed to manipulate users using urgency, fear, and fake links. By analyzing email content, links, and headers using tools like VirusTotal, MXToolbox, and Google Header Analyzer, it is possible to detect phishing attempts. Awareness and careful verification are essential to prevent such attacks.